8 Self-winding watches

8.1 Historical background

8.1.1 The self-winding pocket watch

The first self-winding watches were made in 1770. This invention is attributed to Abraham-Louis Perrelet from Le Locle, who was born in 1729. At this time the only portable timepiece was the pocket watch.

Perrelet's idea was to use the **energy created by the wearer's movements** to wind the watch, i.e. the mainspring, through the ascending and descending movement of an oscillating weight linked to the winding mechanism; such watches were called *perpetual watches.*

The majority of self-winding systems in watches of this period were based on the same principle *(fig. 8-2)*: when it ascends the oscillating weight ***(1)*** carries the winding spring ***(2)*** which is fixed to it. This spring transmits the movement to the winding wheel ***(3)***. The stop spring ***(4)*** prevents the winding wheel from turning backwards; (the winding mechanism turns in only one direction). The buffer pin ***(5)*** restricts the movement of the oscillating weight and the spring ***(6)*** returns it to its original position. The rotation of the winding wheel is transmitted to the barrel arbor through a reduction gear train. When the mainspring is fully wound, the pin ***(7)*** lifts the stop spring preventing the mainspring being overwound, which might damage certain parts of the mechanism.

fig. 8-1 : A watch made by Abraham-Louis Breguet with a self-winding mechanism (1794).

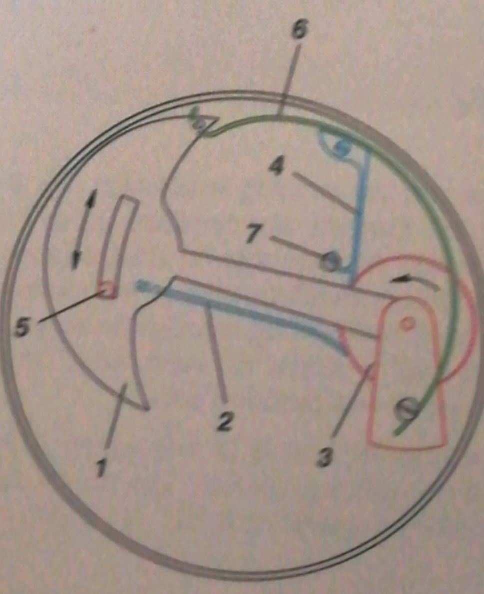


fig. 8-2

One of the main problems of such a mechanism was the noise made by the oscillating weight knocking on the buffer pin, which was unpleasant for the wearer. In addition, this shock caused excessive wear on the parts.

Several years later a watch was discovered which was also thought to have been made by Abraham-Louis Perrelet. It was fitted with a mechanism which is similar to those used today *(fig. 8-3)*. In effect, this watch has a *rotor*-type oscillating weight which pivots in the centre of the movement and has no buffer pin to restrict its movement. Furthermore, both directions of rotation are used to wind the mainspring, thanks to a system which includes two pawl wheels.

fig. 8-3

As a pocket watch, this early model had a serious inconvenience, however: a free-moving oscillating weight reacts less well to impulses than a lever suspended horizontally. These impulses, which were mainly vertically downward in nature, have hardly any effect on a watch which is kept in its owner's pocket!

There are only two known examples of this self-winding rotor watch. Most of the self-winding watches made at this time had rocking levers which came in many different shapes and sizes.

During the 18th and 19th centuries only a few hundred self-winding watches were produced.

fig. 8-4

8.1.2 The self-winding wristwatch

The first self-winding wristwatch *(fig. 8-4)* was made by an English watchmaker, John Harwood, who patented his invention in Switzerland in 1923. The oscillating weight pivoted at the centre of the movement. It could swing through an angle of some 130° and had a buffer at each end. The mechanism wound in one direction only.

The special feature of this watch is that it had no manual winding system. The hands could be reset by using the rotating bezel *(fig. 8-5)*.

fig. 8-5

8.2 The oscillating weight

8.2.1 How the oscillating weight functions

The movements of the wearer's arm are transmitted to the mainspring through the **oscillating weight** and the **reduction gear train**. The weight can be fixed at the centre of the movement or alternatively it may be offset.

The self-winding mechanism is generally placed above the movement, on the bridge side.

During the period 1930-1950 rapid progress was made in improving self-winding devices and the rotor model soon achieved its final form.

8.2.2 The different types of oscillating weight

The limited-rotation oscillating weight

As a rule the limited rotation oscillating weight swings through approximately 120°. This path is limited by buffers which are in fact helical damping springs.

The pivot axis is at the centre of the movement *(fig. 8-6)*.

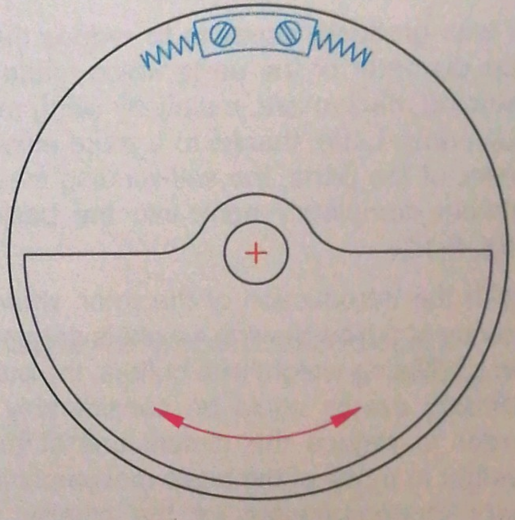


fig. 8-6

The full-rotation oscillating weight

An oscillating weight whose movement is not restricted is generally called a ***rotor***.

The pivot axis is at the centre of the movement *(fig. 8-7)*.

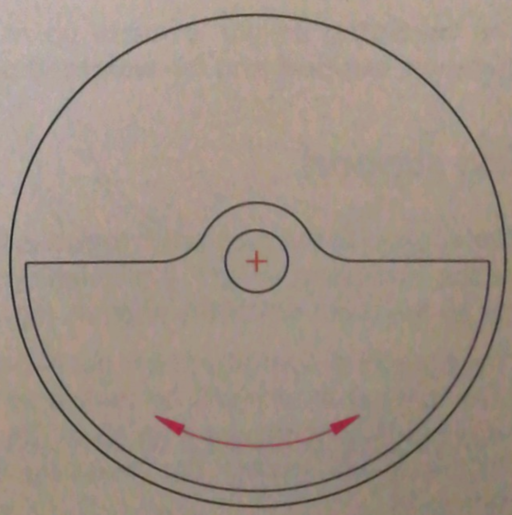


fig. 8-7

The eccentric oscillating weight

In certain cases the pivot of the oscillating weight is not at the centre of the movement.

The weight may be a small-diameter, thick rotor *(fig. 8-8)* or a single-armed lever mounted across the movement, as described in the history of the self-winding watch *(see fig. 8-2)*.

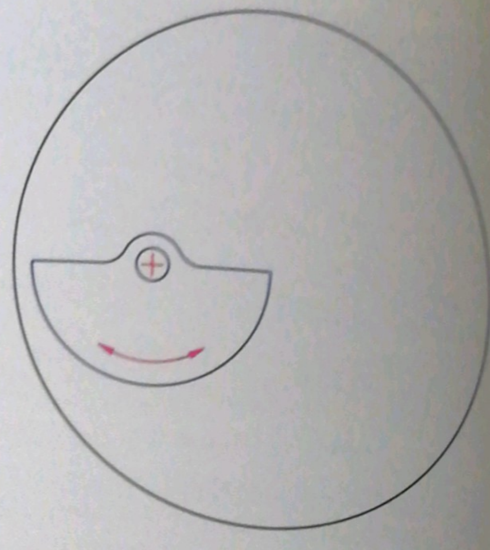


fig. 8-8

8.2.3 The parts which make up the oscillating weight

In the first self-winding watches the height of the movement was more or less doubled by adding a self-winding mechanism *(fig. 8-9)*.

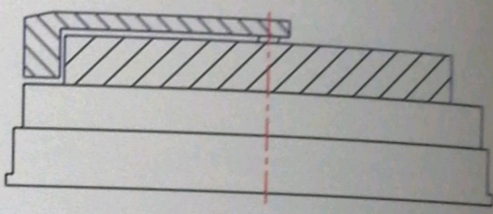


fig. 8-9

It was gradually possible to reduce the height and the diameter of the parts which made up the self-winding mechanism, mainly by using more resistant materials. Later, thanks to a more efficient arrangement of the parts, the self-winding mechanism was almost completely sunk into the basic movement *(fig. 8-10)*.

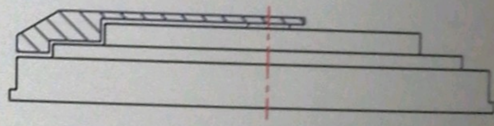


fig. 8-10

With the introduction of the rotor, whereby the efficiency of a free-moving weight is greater than that of an oscillating weight with buffers, the bulk of the self-winding device could be considerably reduced. In order to reduce the dimensions of the oscillating weight to those of the basic movement it was necessary to compensate for the smaller diameter by increasing the weight, so that the moment of inertia remained adequate.

The oscillating weight is made up of two distinct parts: the **support** and the **weighted sector**.

The support

The support is the part which includes the mounting for the oscillating weight; the weighted sector is fitted on the outer rim of the support *(fig. 8-11)*.

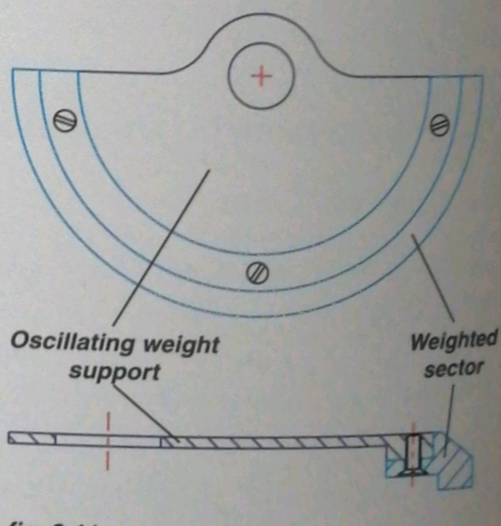


fig. 8-11

The support is normally made out of brass or nickel silver. It must at all costs be flexible so that it is not distorted by any shocks it receives *(fig. 8-12)*; if the support is not flexible the mounting may well be damaged.

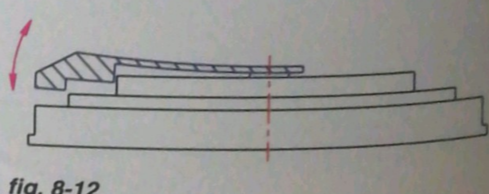


fig. 8-12

The weighted sector

This is the outer, semicircular rim of the oscillating weight. It may be **screwed** *(fig. 8-11)*, **riveted** *(fig. 8-13)*, **inset** *(fig. 8-14)*, or even **soldered** or **cemented** to the support.

The materials used for the weighted sector are mainly:

- ☞ **sintered alloys:** alloys of copper and nickel with a large proportion of tungsten powder,
- ☞ **precious metals:** gold, platinum or iridium.

Remark :

- ☞ lead was used for the first weighted sectors, which were cast into the support *(fig. 8-17)*.

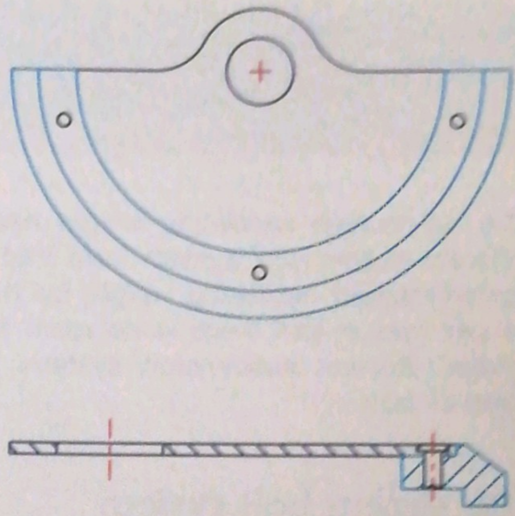


fig. 8-13

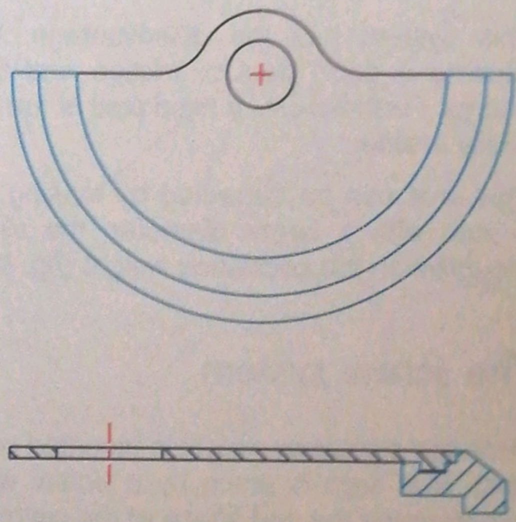


fig. 8-14

8.2.4 The moment of inertia

The moment of inertia of an oscillating weight depends on the radius and thickness of the weight and the material from which it is made.

Two oscillating weights of very different shape may have an identical moment of inertia. In order to have the same moment of inertia an oscillating weight with a small radius must be very thick *(fig. 8-15)* and a very thin oscillating weight must have a large radius *(fig. 8-16)*.

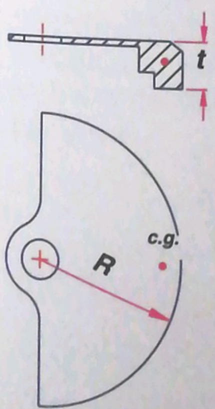


fig. 8-15

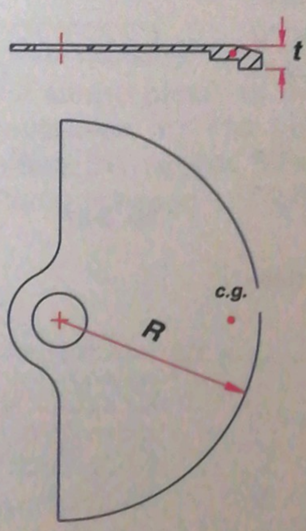


fig. 8-16

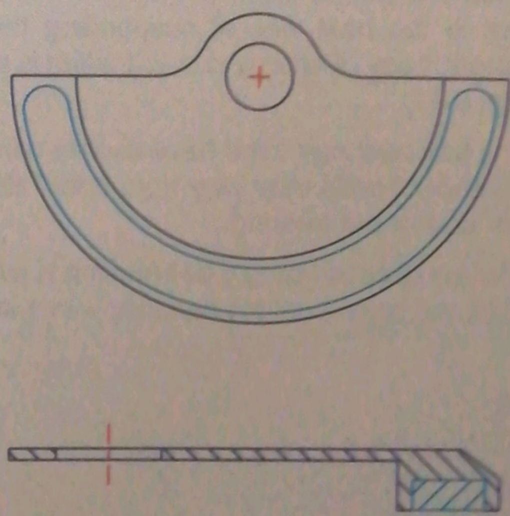


fig. 8-17

8.2.5 The suspension of the oscillating weight

The full-rotation oscillating weight necessitates a different suspension system from that used for a limited-rotation oscillating weight for the plain and simple reason that there is no room for an upper bridge. Several suspension systems have been used to date.

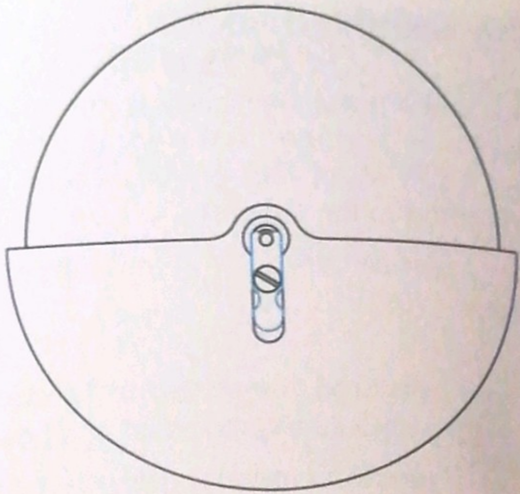


fig. 8-18

The sliding bolt system

One of the simplest systems consists of mounting the rotor on a fixed post and ensuring that it stays in place by a flexible sliding bolt which slips into a groove *(figs. 8-18 and 8-19)*.

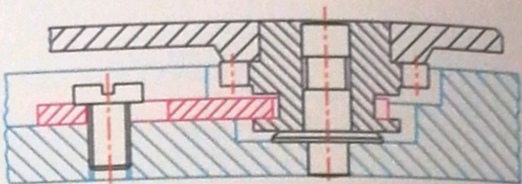


fig. 8-19

This system has the disadvantage, however, of causing a good deal of friction and thus loss of energy. Furthermore, a fixed post is too slender and easily breaks.

This fault can be corrected by making a groove in a post with a larger diameter; the sliding bolt is mounted on the oscillating weight *(fig. 8-20)*.

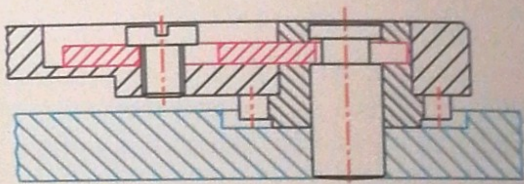


fig. 8-20

The screw system

Here the oscillating weight is mounted on an integral post. It is kept in place by a screw which serves to determine the end shake of the weight *(fig. 8-21)*.

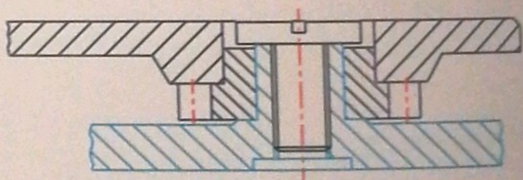


fig. 8-21

The ball-bearing system

From a technical point of view the ball-bearing system is the best way of suspending the oscillating weight. Loss of energy due to friction is virtually zero *(fig. 8-22)*.

The ball bearings used have double-cone races; the number of balls may vary from 5 to 7 depending on the size of the bearing.

The job of assembling a ball bearing is extremely delicate; the balls must roll perfectly within the cage.

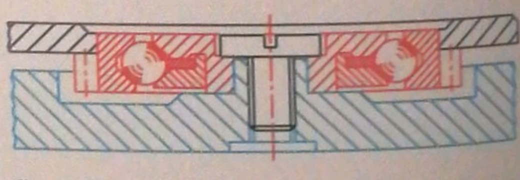


fig. 8-22

8.3 Transmission of the force

8.3.1 The gear train of a self-winding system

The gear train of a self-winding system comprises toothed parts which transmit the force from the oscillating weight to the ratchet, thus winding the mainspring. The gear train is normally made up of the following parts:

1 **Oscillating weight pinion**

2 **Intermediate wheels and pinions (reversers)**

3 **Reduction wheels and pinions**

4 **Ratchet**

Remark :

☞ the ratchet *(4)* is not strictly a part of the gear train of a self-winding mechanism.

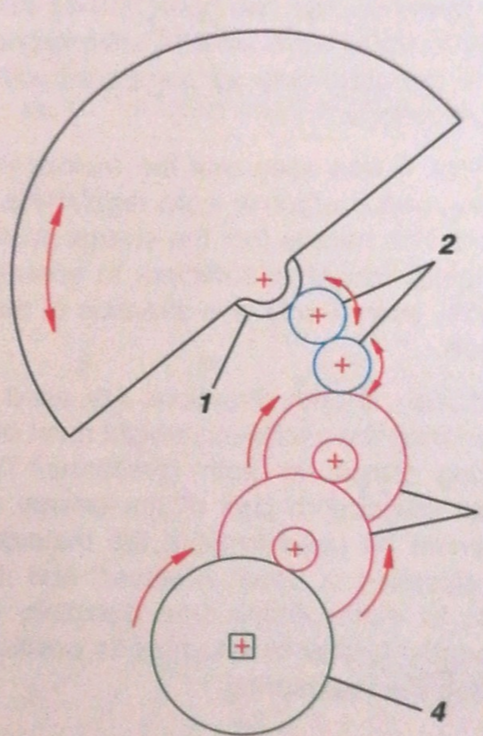


fig. 8-23

As a rule the gear train of a self-winding mechanism *(figs. 8-23 and 8-24)* comprises an intermediate train section and a **reduction train** section. The latter serves to **reduce** the initial rate of the oscillating weight and to increase the force used for winding the mainspring.

In order for the mechanism to function well the reduction ratio of the ratchet to the pinion of the oscillating weight must be between **1 : 110** and **1 : 180**. This means that when the ratchet turns once the oscillating weight turns between 110 and 180 times.

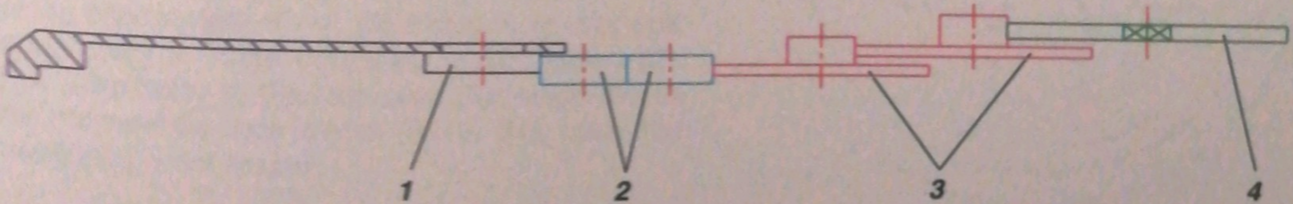


fig. 8-24

8.3.2 Winding in one direction only

The principle of winding in one direction only consists of using the energy produced by the oscillating weight as it moves in one direction. In practical terms this means that the oscillating weight turns freely in one direction while in the other it drives the gear train for the self-winding system, thus winding the mainspring.

After many studies had been carried out it was realised that there were several advantages to winding in one direction only as compared with winding in both directions.

In effect, it was seen that the mainspring of a self-winding watch which is worn regularly is always fully wound. This means that the energy produced by the oscillating weight is sufficient to ensure the power reserve, even if only one direction of the movement is used.

In addition, if both directions are used to wind the mainspring the oscillating weight must overcome the winding torque in both directions. This winding torque will absorb part of the energy which could otherwise be used to wind the mainspring. If only one direction is used, however, and if the weight starts to move in the free direction, it will move noticeably further to return to its position of rest, in winding the mainspring.

Remark :

- the position of rest means that the centre of gravity and the centre of rotation of the oscillating weight are vertically aligned.

Finally, the principle of winding in one direction only means that the mechanism is relatively simple and **is made up of few parts.**

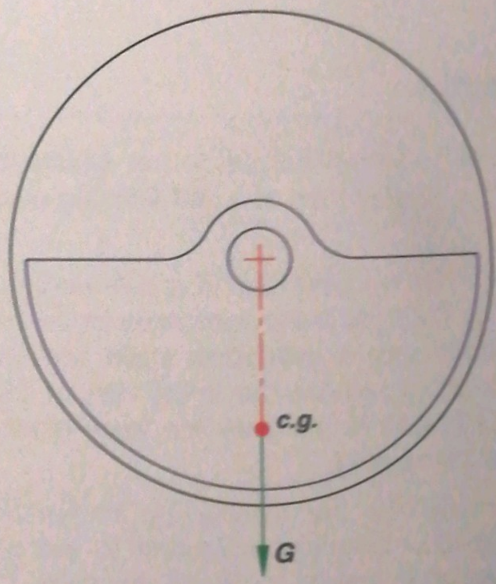


fig. 8-25 : Position of rest.

Mechanism with a limited-rotation oscillating weight *(figs. 8-26 and 8-27)*

In the mechanism illustrated here a free wheel ***(6)*** is mounted on the axis of the oscillating weight ***(1)***. A pawl ***(2)*** is fixed to the weight and causes the free wheel to turn, through its *ratchet teeth*, in the direction shown.

The pawl spring ***(3)*** ensures that the pawl is constantly pressing on the free wheel. Through the action of its spring ***(5)***, a second pawl ***(4)*** acts as a retainer.

The free wheel ***(6)*** has a pinion ***(7)***, which drives the reduction gear train and its wheels and pinions ***(8 and 9)***. The driving force is then transmitted to the ratchet ***(10)***. The pawl ***(11)*** and its spring ***(12)*** prevent the ratchet from turning backwards.

When the oscillating weight turns in the other direction the pawl ***(2)***, which is fixed to it, unclicks from the free wheel without driving it. The mainspring is not wound. The free wheel is thereby held still by the second pawl ***(4)*** through the action of its spring ***(5)***.

The springs ***(13 and 14)*** serve to damp the movement of the oscillating weight.

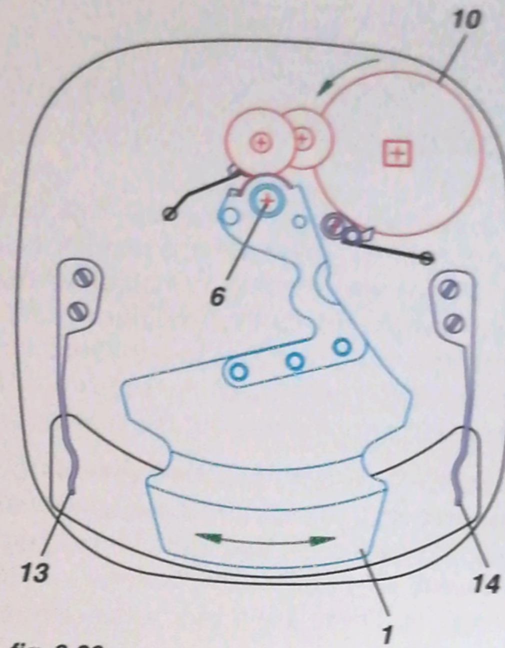


fig. 8-26

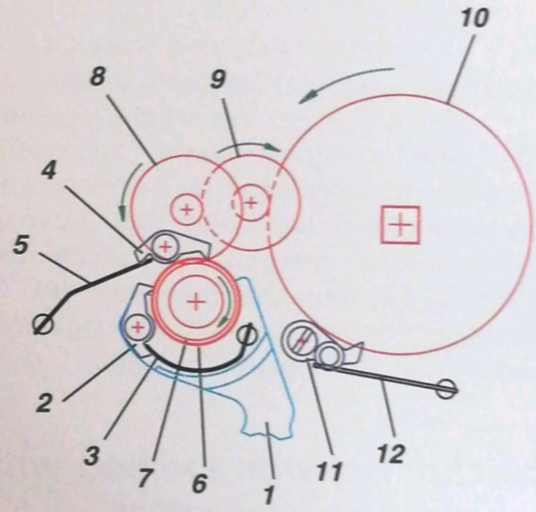


fig. 8-27

Mechanism with a full-rotation oscillating weight *(fig. 8-28)*

The oscillating weight ***(1)*** is screwed on to the hub of a ball bearing which carries the oscillating weight pinion ***(2)***. This pinion is in gear with a sliding intermediate wheel ***(3)***, which sits in a slot and is itself in gear with the intermediate stop wheel ***(4)***, the latter being controlled by a stop pawl ***(5)***. A spring ***(6)*** releases the sliding wheel and the pawl.

The intermediate stop wheel drives the reduction gear train which transmits the driving force to the ratchet ***(9)*** through two reduction wheels and pinions ***(7 and 8)***.

When the oscillating weight turns clockwise it winds the mainspring.

When the oscillating weight turns anticlockwise the sliding intermediate wheel ***(3)*** slides in its slot and moves away from the oscillating weight pinion ***(2)***. This is facilitated by the stop pawl ***(5)***, which blocks the intermediate stop wheel ***(4)***. In this case the mainspring is not wound.

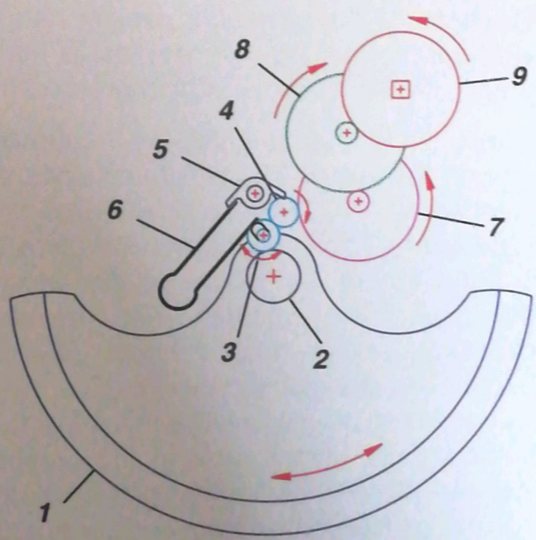


fig. 8-28

8.3.3 Winding in both directions

In a system which winds the mainspring in both directions the alternative movements of the rotor have to be transformed into a continuous rotation in order to wind the spring. In other words, whether the oscillating weight rotates in one direction or the other the reduction gear train must turn only in the direction necessary for winding the mainspring.

It is only by inserting a mechanical device called a ***reverser*** in between the oscillating weight and the reducing gear-train that the self-winding mechanism can work whichever direction the oscillating weight is rotating. There are three types of reversers.

Cam or pawl-lever mechanism

As it swings, the oscillating weight ***(1)*** causes the pawl lever ***(3)*** to jerk because of the eccentric cam ***(2)*** which is fixed to the weight. This movement is transmitted to the two pawls - the leading pawl ***(4)*** and the retaining pawl ***(5)*** - which are under the pressure of their spring ***(6)***. They serve to push and lock the wheel and pinion ***(7)***, which drives the first wheel of the reduction gear train ***(8)***. The force is subsequently transmitted to the wheel and pinion ***(9)*** and thus to the ratchet *(fig. 8-29)*.

Mechanism with toothed wheels which gear alternately

Among the different mechanisms by which both directions of the oscillating weight can be used to wind the mainspring, the simplest is that of change gears, of which there exist numerous different types.

The oldest is without any doubt the **simple reverser** *(figs. 8-30 and 8-31)*, which consists of an intermediate wheel ***(3)*** which pivots on a rocking lever ***(4)***, fixed to either a bridge or the plate, and always in engagement with the wheel ***(2)***, which is fixed to the oscillating weight ***(1)***. This intermediate wheel gears with one wheel or the other ***(5 and 6)***, turning in opposite directions, depending on the direction of the rotation of the oscillating weight. These two wheels must always turn in the correct direction; a pawl ***(8)*** and its spring ***(9)*** prevent them from reversing. The force is subsequently transmitted to the wheel and pinion ***(7)*** and thus to the ratchet.

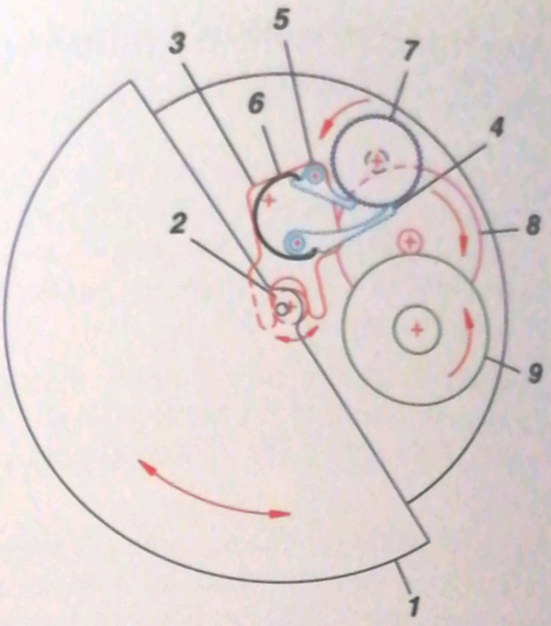


fig. 8-29

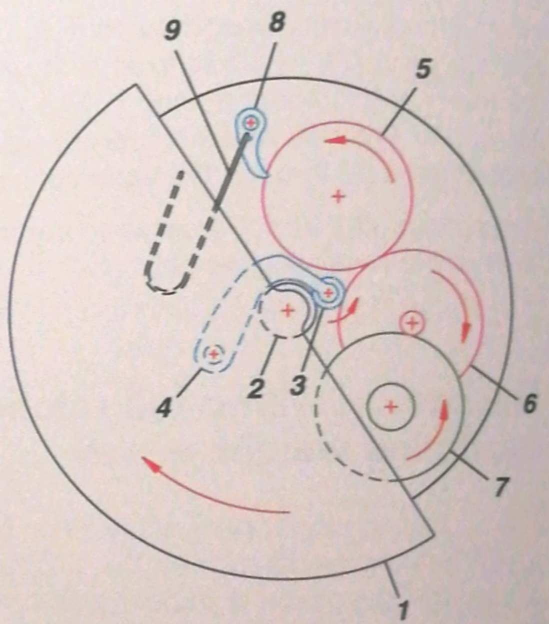


fig. 8-30

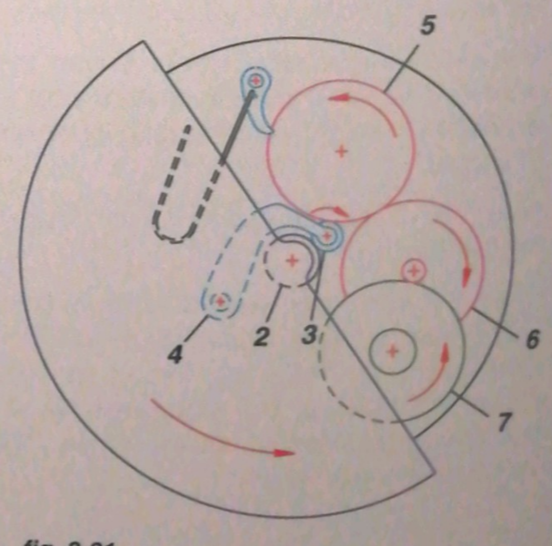


fig. 8-31

Another common system involves **two simple reversers** *(fig. 8-32)*. Whichever direction the oscillating weight ***(1)*** rotates the wheels and pinions ***(6 and 7)*** must turn in the direction indicated (in this case). In order to ensure that they do so, two intermediate wheels are used, which are constantly in mesh ***(3 and 4)*** and pivot on a small rocking platform ***(5)***.

When the oscillating weight turns clockwise, the platform pivots through the moment of force exerted on the intermediate wheel ***(3)***, and it is the latter which gears with the wheel and pinion ***(6)***.

When the oscillating weight rotates anticlockwise, *(fig. 8-33)* the platform pivots through the moment of force exerted on the intermediate wheel ***(3)***, and it is the intermediate wheel ***(4)*** which gears with the wheel and pinion ***(6)***.

Through its spring ***(9)*** the pawl ***(8)*** ensures that the wheel and pinion ***(6)*** are correctly positioned and do not reverse.

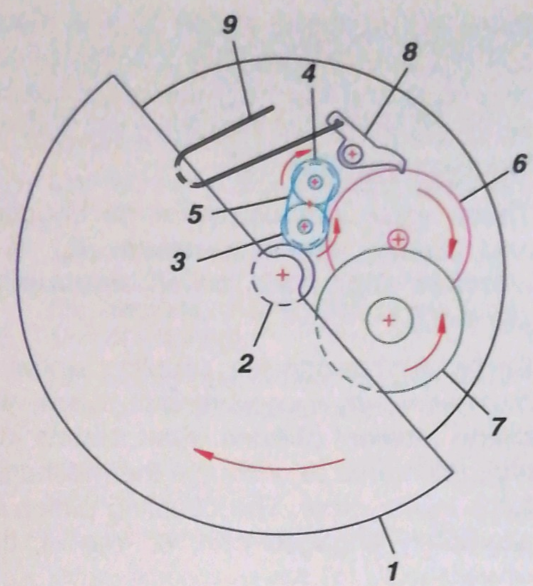


fig. 8-32

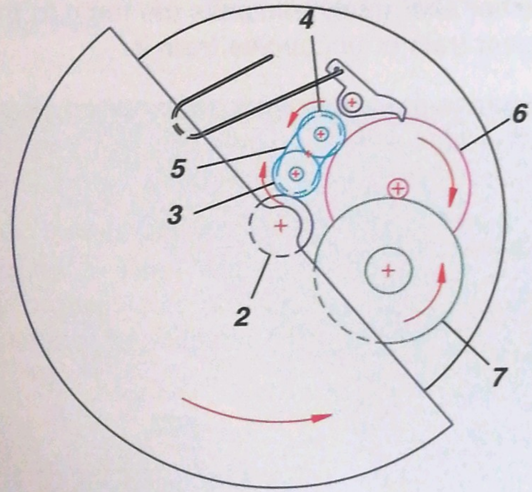


fig. 8-33

Mechanism with a wheel and pinion coupling

In this case two mobiles of the same type gear together. They are each made up of the **coupling wheel**, the **coupling pinion** with normal teeth and ratchet teeth, and one or several **pawls** fixed to the coupling wheel *(see fig. 8-36)*.

If the oscillating weight ***(1)*** turns clockwise *(fig. 8-34)* the pinion ***(2)***, which is fixed to it, drives the first coupling wheel ***(3)***. The pawl on this wheel drives the coupling pinion ***(4)*** via the ratchet teeth.

The pinion then transmits the force to the first wheel of the reduction gear train ***(9)*** which must turn in the direction indicated.

This drives the second coupling wheel ***(6)***, which meshes with the first, and its pawl ***(8)*** rides over the ratchet teeth of the coupling pinion ***(7)*** without driving the latter (which is then free) and follows the movement of the first wheel of the reduction gear train ***(9)***.

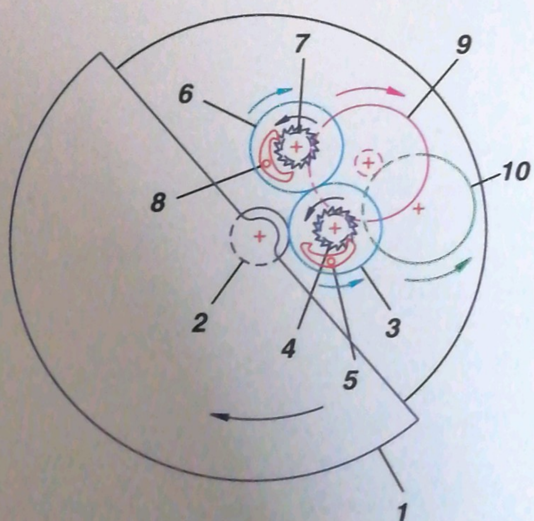


fig. 8-34

Remark :

- the coupling wheel and pinion may have several pawls *(see fig. 8-36)*; they serve the same purpose as the pawl described above.

When the oscillating weight turns in the other direction *(fig. 8-35)* it is the first coupling wheel ***(3)*** which uncouples and it is the pinion of the second coupling wheel ***(7)*** which will then drive the first wheel of the reduction gear train ***(9)***.

There exist various types of coupling wheels and pinions: involving **pawls** *(fig. 8-36)*, **roller clutches** *(fig. 8-38)* or an **uncoupling spring** *(fig. 8-37)*.

By its very design the coupling wheel and pinion must serve as a conventional mobile which transmits a moment of force when it turns in one direction, and serve to uncouple the mechanism when it turns in the other. The coupling pinion is therefore alternately engaged with, or free of, the coupling wheel.

By combining two coupling wheels and pinions of this type it is easy to see that, depending on the direction of rotation of the oscillating weight, one or other alternately transmits the force to the reduction gear train or uncouples from it.

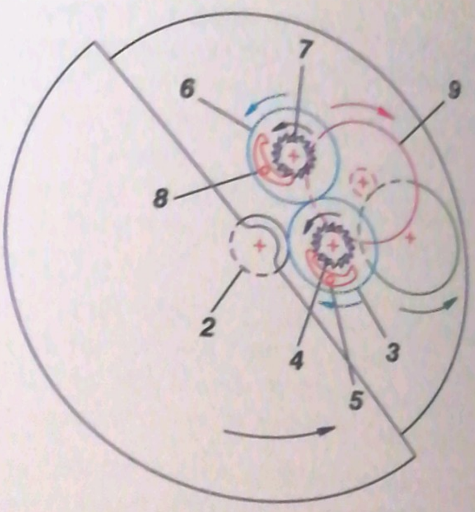


fig. 8-35

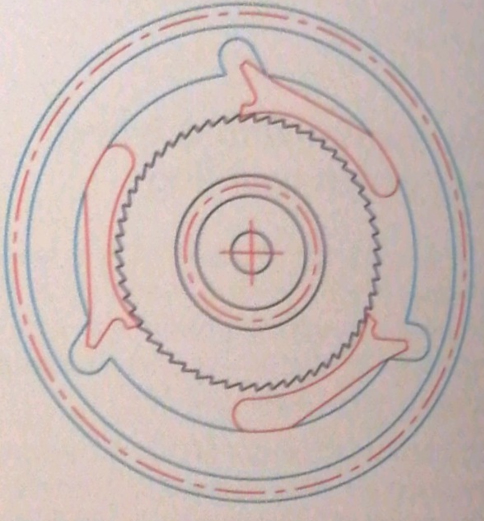
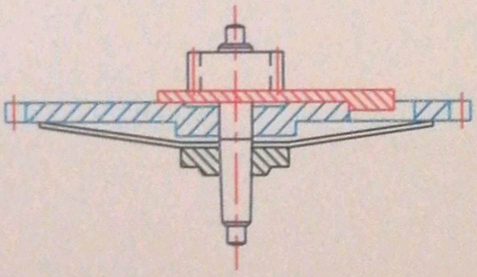


fig. 8-36

Remark :

☞ self-winding mechanisms which have reversers in the form of *coupling wheels and pinions* do not have a pawl which controls the direction of rotation.



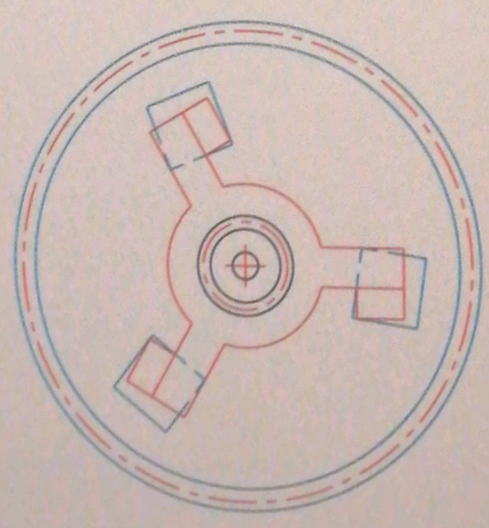


fig. 8-37

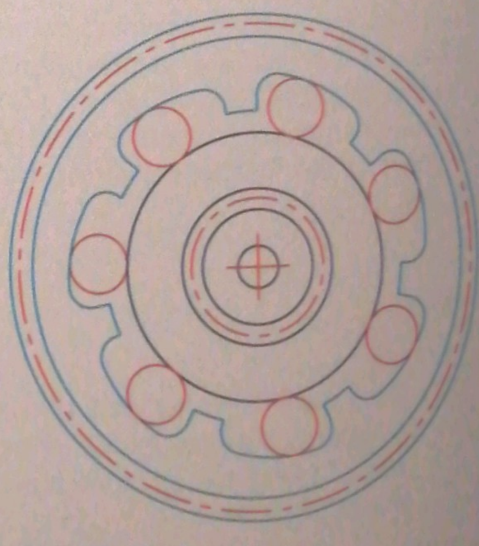


fig. 8-38

8.3.4 Requirements regarding the function of reversers

The requirements for a mechanical reverser are quite stringent. Each year **a reverser changes the direction of movement between 10^6 and 10^7 times !**

The rate at which the force acts and then stops, as well as the intensity of the force may vary considerably. The reverser plays a vital role with respect to the quality of the self-winding mechanism.

Reversers in self-winding watches must fulfil the requirements listed opposite *(fig. 8-39)*.

- Long lifespan for continual functioning without any maintenance.
- Ability to function properly when subjected to shock.
- Low moment of release when direction of rotation changes.
- Construction which does not allow for any modification of the way it functions (for example, modification of the tension of a spring).
- High idling rate without causing wear (this condition applies only to mechanisms with coupling wheels and pinions which are not uncoupled for manual winding).
- Low moment of rotation when idling.
- Small dimensions and flat silhouette.
- Simple design, few parts.
- Lowest price possible for a high level of reliability.

fig. 8-39

fig. 8-40 : "El Primero" self-winding chronograph movement manufactured by Zenith.

8.3.5 The uncoupling mechanism in the manual winding mode

All self-winding watches also have a manual winding system with a winding stem. This is essential even if it is rarely used.

In certain models the self-winding system drives the manual winding mechanism, which idles as far as the winding pinion; only the sliding pinion moves away axially. This system causes quite considerable losses through friction, notably of the angle gearing between the crown wheel and the winding pinion, of the sliding pinion on the square of the winding stem, and in particular of the Breguet toothing when it functions. A return-bar spring which is as weak as possible is normally used to reduce these losses.

In order to solve this problem certain winding systems include devices whereby, for example, an auxiliary intermediate wheel ***(3)*** will move away through the tangential force of the ratchet toothing ***(4)***. The crown wheel ***(1)*** drives the intermediate wheel ***(2)***. The pinion which drives the ratchet ***(5)*** is part of the self-winding mechanism *(figs. 8-41 and 8-42)*.

Another device consists of separating the crown wheel ***(1)*** from the ratchet toothing ***(4)*** through a rocking-bar system or by machining the barrel bridge ***(6)*** in such a way that the crown wheel may move laterally *(figs. 8-43 and 8-44)*.

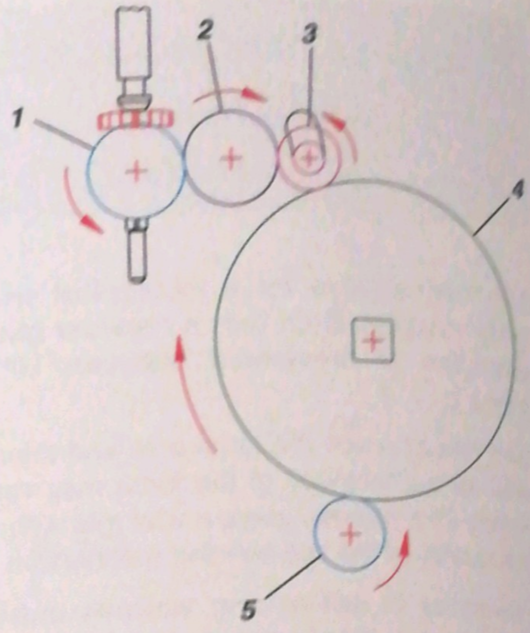


fig. 8-41 : Position for manual winding.

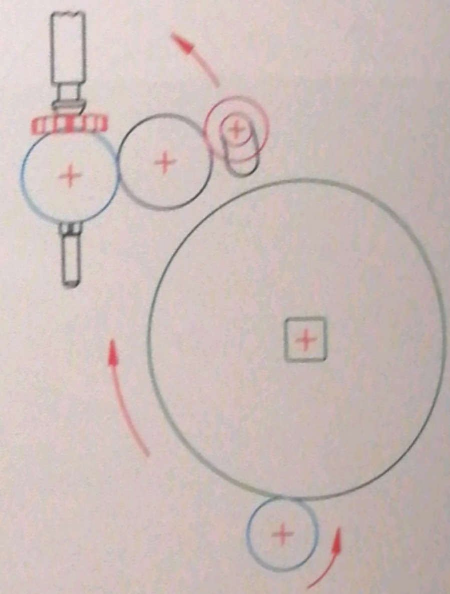


fig. 8-42 : Position for self winding.

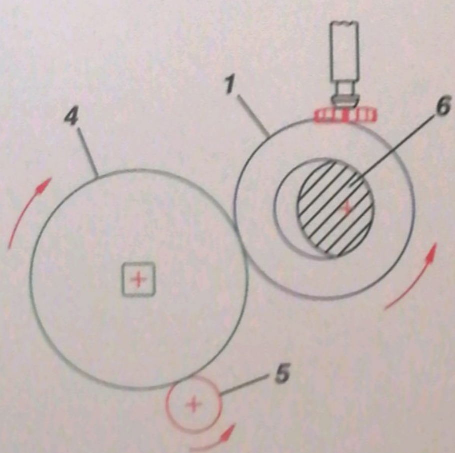


fig. 8-43 : Position for manual winding.

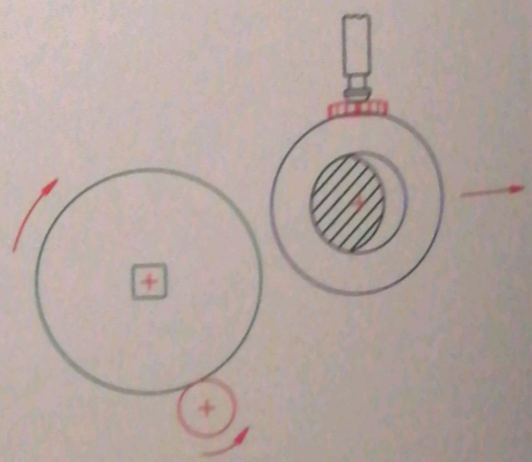


fig. 8-44 : Position for self winding.

8.3.6 The uncoupling mechanism in the self-winding system

When the wearer winds his or her self-winding watch using the winding stem it is important that there is a system which uncouples the reduction gear train in order to avoid the latter rotating too quickly, which would cause excessive wear or even the breakage of one or several parts.

There are various **uncoupling systems** which all work on the same principle: when the watch is wound by the oscillating weight *(fig. 8-45)* the winding wheel and pinion ***(9)*** drive the reduction wheel and pinion ***(8)***, which in turn drive the ratchet-driving wheel ***(7)***. In this position the latter drives the 3-armed spring ***(6)*** which is fixed to the ratchet-driving pinion ***(5)***. The latter acts on the ratchet ***(4)*** and causes the mainspring to be wound.

When the watch is wound manually *(fig. 8-46)* the mechanism linking the manual and the self-winding systems must be uncoupled. This is achieved by the 3-armed spring ***(6)*** which shifts out of the holes in the ratchet-driving wheel ***(7)***, thus uncoupling the system. In this position it is the ratchet which drives the ratchet-driving pinion ***(5)***.

This uncoupling is possible thanks mainly to the action of the stop pawl ***(10)*** which, through its spring ***(11)***, causes the reduction gear train to rotate in one direction only.

The 3-armed spring plays a dual role: it drives (the self-winding system) and uncouples the two systems for manual winding.

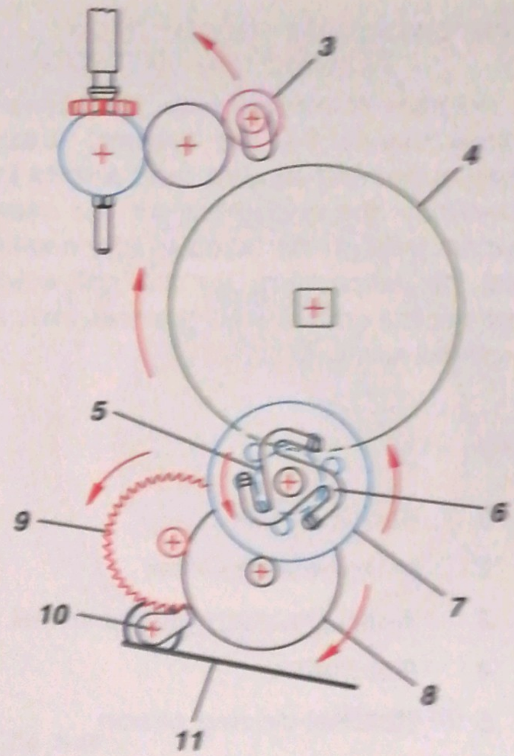


fig. 8-45 : Position for self winding.

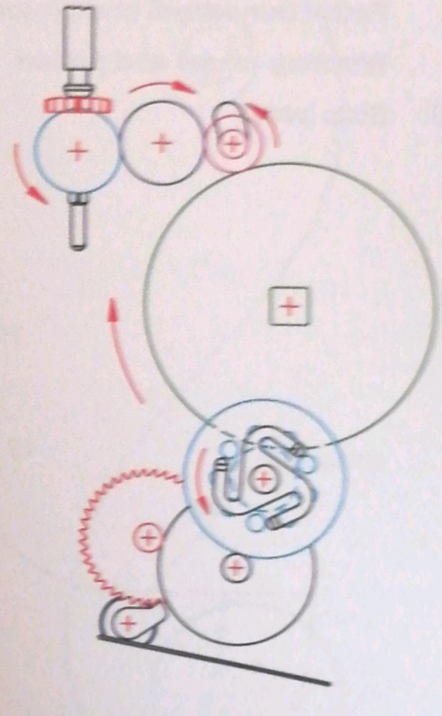


fig. 8-46 : Position for manual winding.

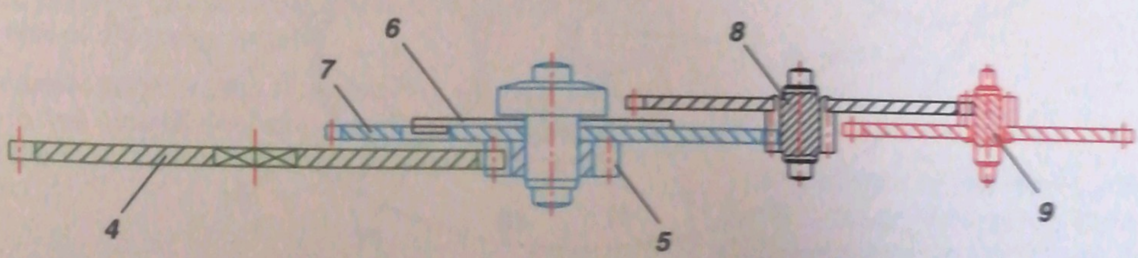


fig. 8-47 : Cross section of system.

The complete mechanism

A diagram of the complete mechanism *(fig. 8-48)* shows clearly the two systems which allow the manual and self-winding mechanisms to work independently: the uncoupling for the manual winding system through the auxiliary intermediate wheel ***(3)*** and the uncoupling for the self-winding system through the ratchet-driving wheel and pinion and its 3-armed spring ***(6)***.

Key

1	**Crown wheel**
2	**Intermediate wheel**
3	**Auxiliary intermediate wheel**
4	**Ratchet**
5	**Ratchet-driving pinion**
6	**3-armed spring**
7	**Ratchet-driving wheel**
8	**Reduction wheel and pinion**
9	**Winding wheel and pinion**
10	**Stop pawl**
11	**Stop-pawl spring**
12	**Reverser**
13	**Reverser**
14	**Mobile platform**
15	**Intermediate pinion**
16	**Pinion fixed to the oscillating weight**
17	**Oscillating weight**
18	**Winding stem**
19	**Winding pinion**

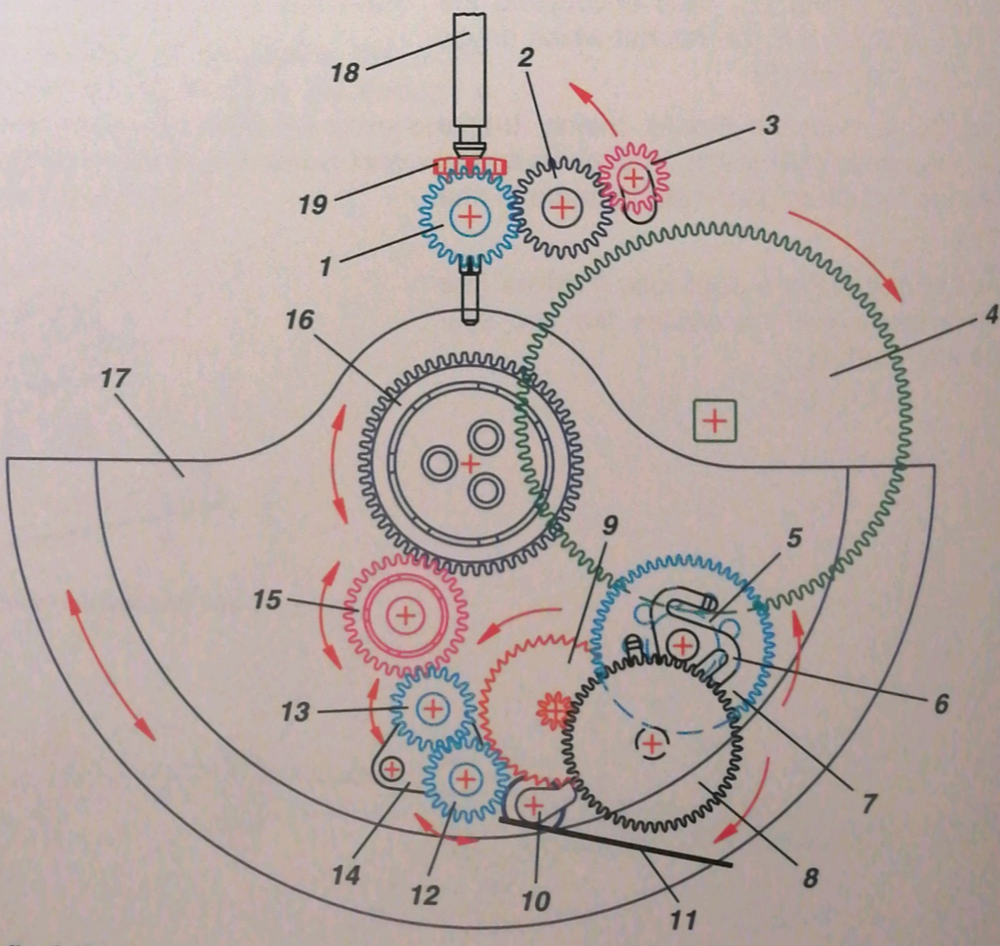


fig. 8-48 : Position for self winding.

8.4 Restricting the self-winding system

8.4.1 Principle

A self-winding watch is being wound constantly. When the mainspring is fully wound it **must not be overstressed** or else it will break. This means that any movement of the oscillating weight must cease to have an effect on the spring when it is already fully wound. If this is not the case, the balance will knock and the self-winding mechanism could be damaged.

In order to ensure that the mainspring is not overstressed it is fitted with a **slip-spring** which acts like a **safety valve**: when the mainspring is fully wound the slip-spring shifts slightly within the barrel drum. This movement must be as brief as possible in order to maintain a maximum amount of torque from the barrel.

The barrel of a self-winding watch does not have a hook on its inner wall. It is smooth or has notches which serve to limit the movement of the slip-spring when it releases the tension on the mainspring; the end of the mainspring stops in each notch in turn. There are 8 notches as a rule and they must be very small (approximately 1 mm wide and 5/100 mm deep) in order not to weaken the barrel wall *(fig. 8-49)*.

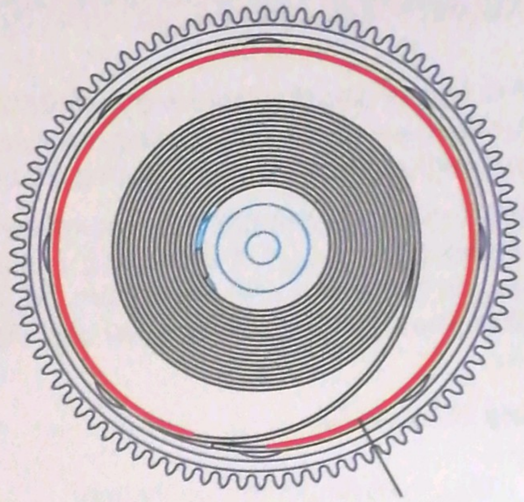


fig. 8-49

8.4.2 The slip-spring

The slip-spring is welded or riveted to the outer end of the mainspring, inside the last coil *(figs. 8-49 and 8-50)*. It presses on this coil and forces it against the inner wall of the barrel drum.

There exists another type of slip-spring which is not fixed to the mainspring and which slides directly along the inner wall of the drum through traction on a hook.

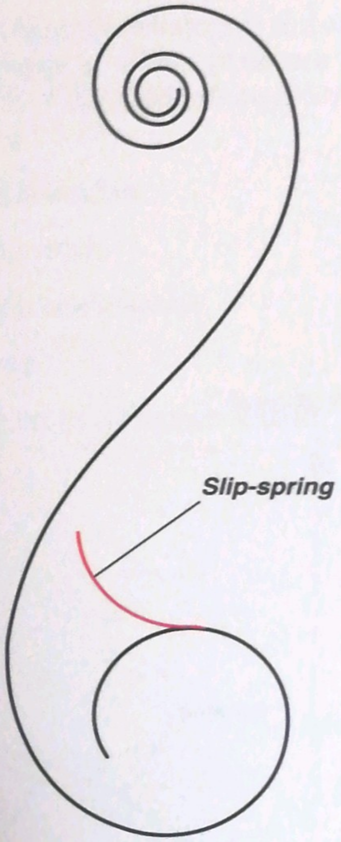


fig. 8-50

The length of the slip-spring is around 10% of the inner circumference of the barrel drum.

8.4.3 Diagram of the spring in a self-winding watch

This type of diagram indicates and allows the excess torque caused by overwinding the mainspring, as well as the effect of the slip-spring, to be measured.

When the mainspring is fully wound the torque is at its maximum *(M_{max})*. If the spring is overwound the M_{max} is exceeded until the torque **(M_{gl})** is reached, when the slip-spring comes into play *(fig. 8-51)*.

Key to formula

M_{max} maximum torque [N·mm]

M_{gl} mean slipping torque [N·mm]

$$\frac{M_{gl}}{M_{max}} \cong 135 - 150\,\%$$

A value below the minimum causes the spring to be released too early, while a value above the maximum may cause the balance to knock.

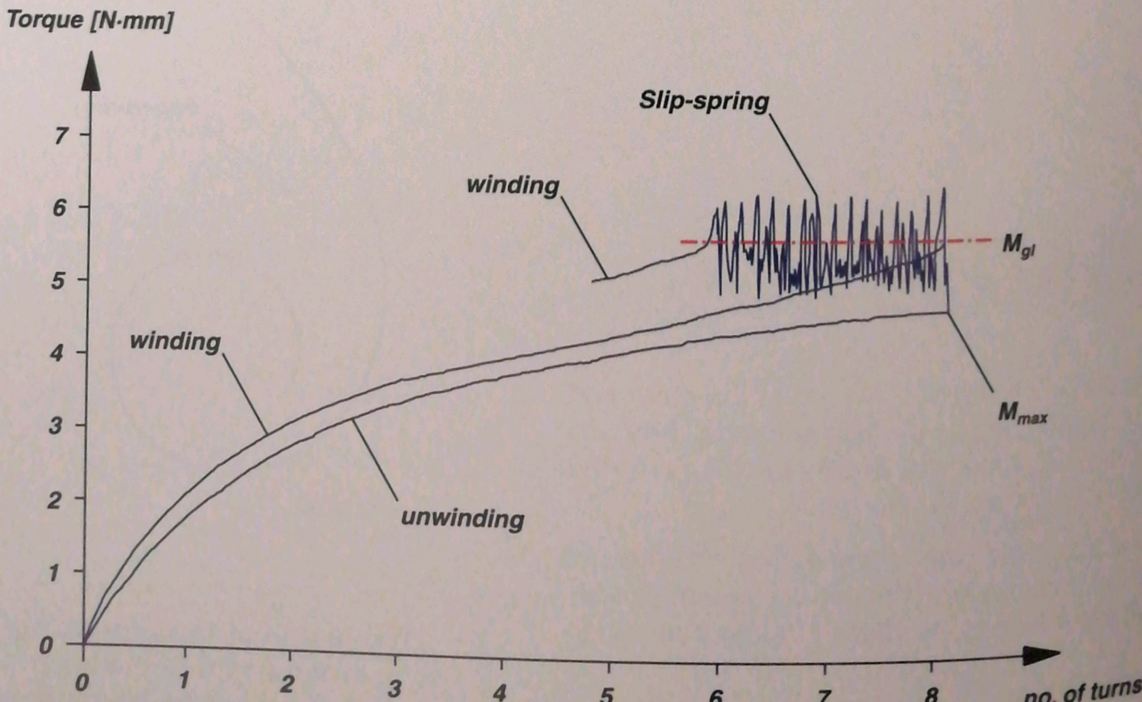


fig. 8-51

Conditions necessary for the slip-spring to function correctly

The action of the slip-spring is very delicate and its correct performance depends on several factors: the shape and tension of the slip-spring, the material used for the barrel drum and the quality of its machining, plating, as well as the lubrication of the barrel and its spring. The manufacturer of the movement will have optimised all these factors through repeated practical testing. It is therefore essential that the corresponding instructions given in technical documentation be followed to the letter.

8.4.4 The winding rate

Practical tests have shown that in a self-winding watch which is worn regularly the power reserve *(see 5.6.2)* is always greater than that of a manually wound watch which is wound at the same time every day.

The winding rate varies according to the temperament of the wearer, his or her degree of activity and state of health, however. The rate is not the same if a given watch is worn by a sedentary person or a physically active person. The terms *weak wearer*, *strong wearer* and *average wearer* are used in this connection.

Tests have been carried out in actual wear and the table opposite *(fig. 8-52)* gives some examples of rates according to the type of activity of the wearer. It is interesting to note that some very simple movements may have a major influence on the self-winding system.

Activity	**Winding rate**
Sleeping	1.2
Driving a car	3.4
Walking	8
Having breakfast	13
Morning wash	19
Washing one's hands	41
Dressing	63
Putting on or taking off a coat	92

fig. 8-52

Instruments for checking the winding rate

Tests in actual wear are complicated and often provide more information concerning the wearer than the watch itself ! Furthermore, such tests must be carried out on a large number of people if the results are to be considered representative.

It is therefore necessary to be able to measure the winding rate objectively and, in order to obtain analogous results, testing must always be done under the same conditions. This is only possible through **mechanical testing**.

The testing apparatus subjects the watch to conditions which are as close as possible to actual wear.

There exist various different types of instruments for measuring the efficiency of self-winding mechanisms *(fig. 8-53)*. They generally involve two types of motion: **rotary** and **planetary**.

Using diagrams provided with the instruments it is possible to plot winding rates on a graph depending on the duration of the test (or the number of revolutions of the testing apparatus) and the power reserve of the watch tested *(fig. 8-54)*.

fig. 8-53

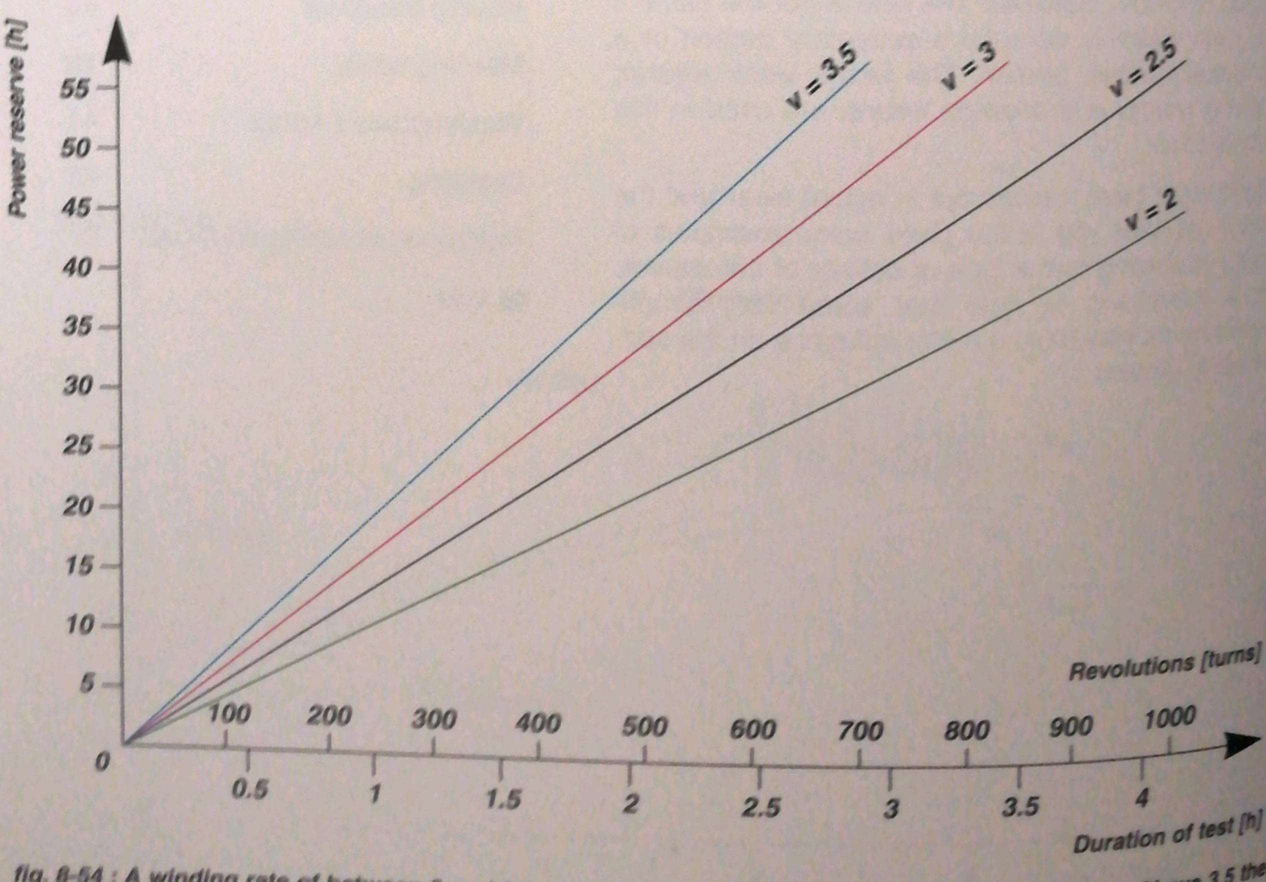


fig. 8-54 : A winding rate of between 2 and 3.5 is considered normal. Below 2 the watch may stop. Above 3.5 the mainspring will be constantly overstressed.